

Koushik Marka

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EDUCATION

SR University

Bachelor of Technology in Computer Science and Engineering

- CGPA: 8.9/10 | Semester Topper — Ranked 1st in CSE Department, 3rd Year

Hyderabad, Telangana

Jun 2024 – May 2028

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), Java, C++, Go (familiar)

Systems & Algorithms: Distributed Systems, DAG/Topological Sort, Graph Algorithms, Concurrency, OOP, Software Design Patterns

AI & ML: LLMs, Transformer Models (HuggingFace BART), NLP, RAG, Vector Search, TensorFlow.js, OpenCV

Backend & APIs: Python FastAPI, REST APIs, Pydantic, Streamlit, Node.js

Frontend: React, ReactFlow, Zustand, HTML/CSS, Component Abstraction

Cloud & Infra: Microsoft Azure (AI-900), AWS Cloud Foundations, Oracle AI Vector DB

Tools: Git, GitHub, VS Code, GitHub Copilot, Chrome Extensions API

PROJECTS

Distributed AI Pipeline Orchestrator | *React, ReactFlow, Python, FastAPI, Graph Algorithms* 2025

- Designed and built a full-stack visual pipeline system with 9 composable node types (Input, Output, LLM, API, Condition, Math, etc.), enabling end-to-end orchestration of AI workflows at scale
- Architected a `BaseNode` abstraction layer that decouples node configuration from rendering logic — new node types require only a single config object, eliminating code duplication and improving system extensibility
- Implemented a FastAPI backend with a `/pipelines/parse` endpoint that ingests arbitrary graphs, validates DAG structure using Kahn's topological sort algorithm, and returns node/edge counts with cycle detection in $O(V+E)$
- Built real-time variable parsing in the Text node (`{{variable}}` syntax) that auto-generates ReactFlow handles dynamically, enabling live data-flow wiring without page reloads
- Integrated frontend-to-backend pipeline submission with structured result reporting, supporting iterative debugging of large-scale AI graph configurations

Multi-Modal AI Summarization Platform | *Python, HuggingFace BART, NLTK, pydub, Streamlit* 2025

- Built a scalable, modular NLP system processing 5 input formats (text, PDF, MP3, MP4, WAV) using an OOP architecture that cleanly separates ingestion, preprocessing, and inference layers
- Engineered a dual-stage NLP pipeline: extractive summarization via NLTK sentence scoring followed by abstractive summarization using a HuggingFace BART transformer, improving output quality over single-method approaches
- Implemented audio preprocessing with pydub silence detection prior to speech-to-text transcription, reducing noise-induced transcription errors and improving downstream summarization accuracy
- Deployed as a full-stack interactive web application via Streamlit, supporting real-time upload, transcription, and summarization for end users

AI Traffic Signal Management System | *Python, OpenCV, NumPy, Streamlit* 2025

- Developed a real-time computer vision system concurrently processing 4 live video lanes to detect vehicles and dynamically adjust traffic signal timing — **selected among Top 50 teams at Smart India Hackathon (SIH)**
- Designed a modular, production-ready architecture decoupling vehicle detection logic from dashboard rendering, enabling independent scaling and testing of each subsystem
- Delivered as part of a 6-member agile team, coordinating parallel workstreams across CV pipeline, backend logic, and UI within a time-constrained hackathon environment

Memory Helper (Browser AI Inference Engine) | *JavaScript, TensorFlow.js, face-api.js, Chrome APIs* 2025

- Built an end-to-end ML inference pipeline running entirely at the browser layer using TensorFlow.js and face-api.js — no server dependency, enabling offline-capable real-time face tracking and speech integration
- Integrated Web Speech API with anti-spam cooldown logic and built a secure caregiver dashboard with full CRUD backed by `chrome.storage.local`, ensuring privacy-preserving local-only data storage

ACHIEVEMENTS & ACTIVITIES

Smart India Hackathon (SIH) – Internal Round SR University, 2025
Selected among Top 50 teams university-wide for AI Traffic Signal Management System

Semester Topper SR University, 2025
Ranked 1st in CSE Department; currently in 3rd Year with CGPA 8.9/10

VS Code Dev Days 2025 Microsoft, Hyderabad
Selected attendee; explored GitHub Copilot, AI-assisted development, and modern collaboration workflows

CSI – Computer Society of India, SR University Chapter 2024 – Present
Active member; helped organize technical workshops, coding events, and peer learning sessions

CERTIFICATIONS

Microsoft Azure AI Fundamentals (AI-900) Microsoft, 2025
Verify Credential

Oracle AI Vector Search Certified Professional Oracle, Oct 2025
Verify Credential — RAG, Vector Embeddings, HNSW/IVF Indexes

AWS Academy Graduate – Cloud Foundations Amazon Web Services, Oct 2025
Verify Credential